LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to circuit breakers and various protection provided by circuit breakers
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to circuit breakers and various protection provided by circuit breakers.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of CB operation and inspection; Safety: Do not reset blown CBs without fault isolation; Supervision: Instructor verifies fault identification before reset.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a circuit breaker in an aircraft?
(b) List three common causes of aircraft wiring degradation.
(c) Why are circuit breakers preferred over fuses in modern aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic construction of circuit breakers • Types of circuit breakers used in aircraft systems • Mechanism of thermal and magnetic tripping • Standard locations and identification of CBs • Visual inspection criteria for CB integrity	L & Ds	PPT, IB	15 min
(b)	• Causes of wiring degradation including chafing and corrosion • Impact of degraded wiring on circuit protection • Relationship between wire condition and CB operation • Common failure modes in aircraft electrical systems • Preventive maintenance measures for wiring integrity	L & Ds	PPT, PDF	20 min
(c)	• Procedures for resetting circuit breakers safely • Troubleshooting steps after CB trip • Documentation requirements for CB resets • When to replace a circuit breaker permanently • Coordination with other protection systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a degraded wire can cause a circuit breaker to trip unnecessarily.
(b) Describe the correct procedure for investigating a tripped circuit breaker before resetting it.
(c) Identify two signs of wiring degradation that require immediate maintenance action.

12. Home Assignment and Follow Up	Review the aircraft wiring diagram for your assigned station and identify three circuit breakers; prepare a brief note on their associated systems and potential degradation risks to discuss in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Determination of CB size and maximum continuous current flow through CBs.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Instructor, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min